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CLINICAL RESEARCH ARTICLE

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Well-being comparisons and mental health outcomes following loss: a two-wave longitudinal study

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ABSTRACT

Background: The loss of a significant other may lead to prolonged grief disorder (PGD), posttraumatic stress disorder (PTSD), depression, anxiety, somatic symptoms, and loneliness. Comparative thinking about one's well-being (e.g. social, temporal, counterfactual, or expectation-based comparisons) may act as a cognitive link between bereavement and psychopathology. For example, frequently comparing current well-being to pre-loss levels (i.e. past temporal comparison) and perceiving a gap (comparison discrepancy) can intensify negative emotions (comparison affective impact), which may, in turn, exacerbate psychopathology. We hypothesized that aversive well-being comparison frequency, comparison discrepancy, and negative comparison affective impact would each predict higher psychopathology over time, with comparison discrepancy and affective impact sequentially mediating the association between comparison frequency and psychopathology.
Methods: Our two-wave longitudinal study of bereaved individuals ($N = 315$) assessed well-being comparisons, worry, rumination, avoidance, childhood adversity, and mental health outcomes over six months post-loss.
Results: Preregistered hypotheses were partly confirmed. Higher frequency of aversive well-being comparison, comparison discrepancy, and negative comparison affective impact predicted elevated psychopathology (PGD, PTSD, depression, anxiety, somatic symptoms, & loneliness) symptoms six months later. Nuanced effects persisted after accounting for autoregressive pathways, worry, rumination, avoidance, and childhood adversity. In mediation pathways, frequent comparison amplified perceived discrepancies, eliciting negative affective impact, which in turn contributed to depression, anxiety, and somatic symptoms.
Conclusion: Our work suggests that well-being comparison plays a significant role in mental health outcomes following loss.

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Comparative thinking; social comparison; prolonged grief; rumination; childhood adversity

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Pensamiento comparativo; duelo prolongado; rumiación; adversas en la infancia

HIGHLIGHTS

- Aversive well-being comparison predicted elevated psychopathology six months post-loss.
- Well-being comparison may be a key cognitive mechanism linking loss to psychopathology.
- Interventions should address maladaptive well-being comparison after loss.

Comparaciones de bienestar y resultados de salud mental tras una pérdida: un estudio longitudinal de dos olas

Antecedentes: La pérdida de un ser querido puede dar lugar al trastorno de duelo prolongado (TDP), al trastorno de estrés postraumático (TEPT), así como a depresión, ansiedad, síntomas somáticos y soledad. El pensamiento comparativo sobre el propio bienestar (por ejemplo, comparaciones sociales, temporales, contrafácticas o basadas en expectativas) puede actuar como un vínculo cognitivo entre el duelo y la psicopatología. Por ejemplo, comparar con frecuencia el bienestar actual con los niveles previos a la pérdida (es decir, una comparación temporal con el pasado) y percibir una diferencia entre ambos (discrepancia en la comparación) puede intensificar las emociones negativas (impacto afectivo de la comparación), lo que, a su vez, puede agravar la psicopatología. Planteamos la hipótesis de que la frecuencia de comparaciones aversivas de bienestar, la discrepancia en la comparación y el impacto afectivo negativo de la comparación predecirían, cada uno por separado, niveles más altos de psicopatología a lo largo del tiempo. Asimismo, se esperaba que la discrepancia en la comparación y el impacto afectivo actuaran secuencialmente como mediadores de la relación entre la frecuencia de las comparaciones y la psicopatología.
Método: Nuestro estudio longitudinal de dos olas con personas en duelo ($N = 315$) evaluó las comparaciones de bienestar, la preocupación, la rumiación, la evitación, las experiencias adversas en la infancia y los resultados de salud mental durante los seis meses posteriores a la pérdida.
Resultados: Las hipótesis preregistradas se confirmaron parcialmente. Una mayor frecuencia de comparaciones aversivas de bienestar, la discrepancia en la comparación y el impacto afectivo negativo de la comparación predijeron niveles más elevados de síntomas psicopatológicos (TDP, TEPT, depresión, ansiedad, síntomas somáticos y soledad) seis meses después. Se observaron efectos matizados que se mantuvieron incluso al considerar las trayectorias autorregresivas, la preocupación, la rumiación, la evitación y la adversidad en la

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infancia. En las rutas de mediación, la frecuencia de las comparaciones intensificó las discrepancias percibidas, lo que provocó un impacto afectivo negativo, el cual a su vez contribuyó a la depresión, la ansiedad y los síntomas somáticos.
Conclusión: Nuestro trabajo sugiere que las comparaciones del bienestar desempeñan un rol significativo en los resultados de salud mental tras una pérdida.

1. Introduction

Prolonged grief disorder (PGD) is marked by intense yearning for and persistent preoccupation with a deceased loved one, as well as intense emotional pain (Killikelly & Maercker, 2017). The severity and duration of these symptoms, as well as the associated functional impairment, exceed the social, cultural, or religious norms expected within the bereaved person's context (American Psychiatric Association, 2013; World Health Organization, 2018). The diagnosis requires symptoms to persist for at least 6 (ICD-11) or 12 months (DSM-5-TR). PGD has an average estimated prevalence of about 10% to 13% in bereaved individuals (Comtesse et al., 2024; Lundorff et al., 2017). This condition often co-occurs with posttraumatic stress disorder (PTSD), depression, anxiety, somatic symptoms, and increased loneliness (Rueger et al., 2024). PGD's comorbidity is in part informed by symptoms related to identity confusion and re-engagement difficulties (Malgaroli et al., 2018), such as the perception that a part of oneself has died, role uncertainty, difficult pursuing interests and planning for the future (Bellet et al., 2020; O'Connor et al., 2023). As individuals seek to evaluate their current circumstances and anticipate future development, comparison mechanisms may function as an important cognitive link in this process. Yet, there is lack of research on the role of well-being comparison (e.g. social, temporal, or counterfactual comparisons) and mental health outcomes among bereaved individuals. We addressed this research gap by additionally accounting for established transdiagnostic factors, including worry, rumination, avoidance, and childhood adversity (McEvoy et al., 2013; McLaughlin et al., 2012; Morina, 2011).

In the current study, we investigated comparative thinking from a process-oriented perspective (Morina, 2021). In this framework, frequent comparison strengthens the perceived discrepancy between one's own well-being and that of comparison standards, thereby influencing negative or positive emotions (i.e. comparison affective impact), which, in turn, may affect a range of mental health symptoms. This approach contributes to the literature by (a) examining whether comparative thinking processes emerge as a transdiagnostic risk factor beyond worry, rumination, avoidance, and childhood adversity, and (b) identifying potential unique pathways to specific mental health symptoms, such as PGD, PTSD, depression, anxiety, somatic symptoms, and loneliness in the context of loss.

1.2. Comparison types

The assessment of subjective well-being after loss, as any other form of self-judgment, entails comparisons (Vlaev et al., 2011). Well-being comparisons encompass social, temporal, counterfactual, and expectation-based standards (Morina, 2021). *Social comparison* involves evaluating one's well-being relative to others (Festinger, 1954). In the context of bereavement, individuals may frequently evaluate their well-being in comparison to that of other bereaved and non-bereaved individuals. *Temporal comparison* refers to evaluating one's current well-being relative to one's past well-being, or one's anticipated future well-being (Albert, 1977). Bereaved individuals might compare their current well-being to their well-being before the loss, which may reinforce feelings of stagnation or despair. *Counterfactual comparison* is defined as comparing one's current well-being against a mentally simulated alternative state of well-being (i.e. counter to the facts) (Kahneman & Miller, 1986). Bereaved individuals may reflect on how they would be doing if the loss had not occurred, which can amplify feelings of regret, guilt, or a sense of 'what might have been' (Eisma et al., 2021). Last, *expectation-based comparison* implies comparing one's well-being to certain expectations or self-guides, such as how one should be doing at a certain stage of life (Higgins, 1996). In the context of a loss, individuals may evaluate their well-being and grief against their own or societal expectations of how one *should* cope with loss (McLean et al., 2022). If they perceive themselves as failing to meet these expectations, they may experience negative affective and cognitive reactions.

1.3. The process of well-being comparison

Engagement in well-being comparison serves the motives of self-assessment, self-enhancement, self-verification, and self-improvement (Taylor et al., 1995). The general comparative-processing model (gComp; Morina, 2021) provides a theoretical framework that captures the complexities of the comparison process. After selecting a comparison standard, such as social or counterfactual, the individual evaluates

the (dis)similarities between their perception of their well-being and that of the chosen standard. The resulting comparison outcome represents the perceived degree of *discrepancy* between the individual's representation and the standard, leading them to construe their own well-being in relation to the discrepancy. Ultimately, the comparison process may instigate change in the comparer's affect relative to their pre-comparison baseline, defined as *affective impact*. Comparisons can be made relative to upward (better off), lateral (similar), or downward (worse off) standards. Consequently, comparisons can be perceived as aversive, neutral, or appetitive, depending on whether they challenge or align with the comparer's motives (Morina, 2021; Morina & Schlechter, 2023). *Aversive well-being comparisons* in particular have been linked to psychopathology and these include upward social, past temporal, counterfactual, and expectation-based comparisons, as well as downward prospective temporal comparisons (Morina & Schlechter, 2023). To illustrate the comparison process, consider a hypothetical example: Jenny, who recently lost her partner, frequently compares her current well-being to her past well-being and consistently concludes that she now feels worse than she did before. The frequent upward past temporal comparison amplifies the discrepancy between her pre-loss well-being and present well-being, potentially exacerbating her distress and reinforcing a diminished sense of agency, thereby contributing to PGD and other mental symptoms. Individual differences in this process may explain differences in PGD and mental health symptoms.

1.4. Prior research

Although prior research has significantly advanced understanding of and interventions for prolonged grief (LaPlante et al., 2024), there is only limited research on well-being comparison and grief. For example, Eisma et al. (2021) reported that nonreferent upward counterfactual thinking was strongly associated with concurrent prolonged grief and depression symptoms, while self-referent upward counterfactual thinking was linked to these symptoms longitudinally. However, extant literature is limited and has not examined effects of well-being comparison. Indeed, aversive well-being comparison has emerged as an important transdiagnostically relevant factor, with links found to depression, anxiety, PTSD, metacognitions about worries, rumination, and psychological well-being (Morina & Schlechter, 2023; Schlechter et al., 2023; Schlechter, Hellmann, et al., 2024; Schlechter, Hoppen, et al., 2024). However, well-being comparisons have not been examined in bereaved individuals, either in isolation or in conjunction with other established critical risk factors. There is evidence that worry (Akbari & Khanipour, 2018; McEvoy et al., 2013), rumination (Bernstein et al., 2019; Ehring, 2021), avoidance (Eisma & Stroebe, 2017), and childhood adversity (Chen et al., 2020; McLaughlin et al., 2012) represent important risk factors in the context of mental health and grief symptoms.

We examined the incremental value of well-being comparisons by testing whether they predict mental health symptoms independently of worry, rumination, avoidance and childhood adversity or whether their effects diminish when these factors are taken into account. For example, bereaved individuals may engage in well-being comparisons as a coping mechanism for processing their grief, regardless of their predisposition to rumination or excessive worry. Alternatively, a tendency to ruminate or worry could account for the unique variance in well-being comparisons in the context of mental health problems (Schlechter & Morina, 2023). Moreover, by examining multiple mental health outcomes, we can better understand the unique pathways for bereaved individuals. For instance, rumination may be a key predictor of grief and depressive symptoms as repetitive and uncontrollable thoughts about the loss can maintain preoccupation with the deceased, hinder emotional processing, and prolong adjustment to the loss (Ehring, 2021). Worry may primarily predict anxiety (Borkovec et al., 1998), and avoidance and childhood adversity may be primarily linked to PTSD (McLaughlin et al., 2012), to name a few examples of potential unique pathways. Aversive well-being comparison may contribute unique variance to all of these outcomes or may specifically impact certain ones. Exploring these pathways of aversive comparison thinking may increase our understanding of how mental health complaints develop in bereavement. Finally, prior research on bereavement has not scrutinized the components of the comparison process (Morina, 2021), including comparison frequency, discrepancy, and affective impact, which is yet important for a process-based understanding of comparative thinking in mental health, including grief. Our study specifically investigated these elements to better understand their role in prolonged grief and associated complaints. Figure 1 depicts the causal model for the present work.

1.5. The present study

We examined the longitudinal development of mental health complaints in bereaved individuals through two assessments early in bereavement. Specifically, we investigated the development of PGD, PTSD, depressive, anxiety, and somatic symptoms as well as levels of loneliness in individuals who reported loss of a significant other within three months prior to assessment, and re-assessed these same outcomes 6 months later. The second assessment aligns with the time-frame

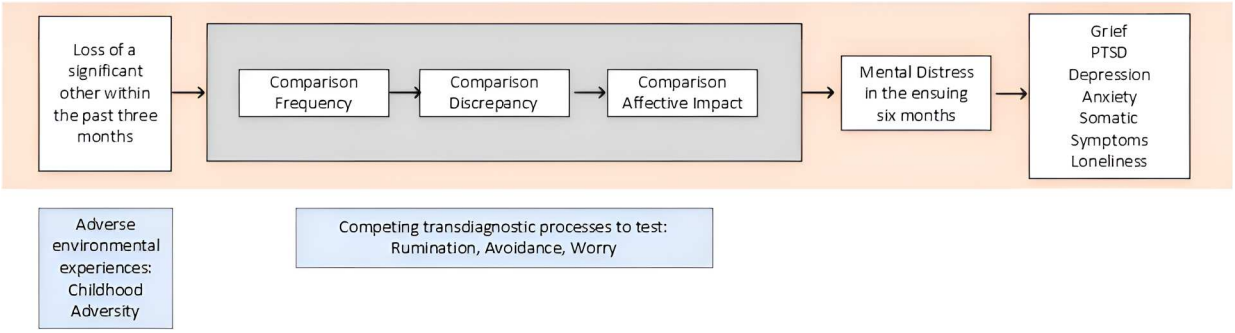


Figure 1. Visual representation of the main model.

specified in the ICD-11 criteria for PGD. We investigated the role of well-being comparisons as a central variable in shaping responses to loss. Our preregistered hypotheses state that all segments of the aversive well-being comparisons process at the first assessment predict higher levels of psychopathological symptoms and loneliness assessed six months later, with this association remaining significant after statistically controlling for worry, rumination, avoidance, and childhood adversity (H1). We did not preregister differential effects on the other assessed mental health outcome variables. Hence, this was investigated exploratorily. Further, we expected mediation effects as formulated in gComp (Morina, 2021), in which high comparison frequency leads to a greater perceived discrepancy (H2), resulting in a more pronounced negative affective impact (H3) and subsequent mental health symptoms (H4).

2. Methods

2.1. Openness and transparency

Our study, including study design, procedures, and our hypotheses, was preregistered https://osf.io/47hpk/?view_only=3d2d9e4e9e8345689d4ab559254ccc93. We note some deviations from our preregistration, which we explain in due detail below. The data for this study are being analyzed for the first time in this manuscript. All data, code, and a codebook for reproducing the results of this study are available on OSF https://osf.io/4kdb9/?view_only=4081bb06e5b34f43bcef4f9d6305200e.

2.2. Procedure and ethics

Participants who experienced a loss within the last three months were surveyed at two time points: (1) from 22 January to 24 January 2024, and (2) from 22 July 22 to 30 July 2024 (i.e. six months later). Participants were recruited via Prolific, a platform for identity-verified individuals aged 18 or older from OECD countries (Palan & Schitter, 2018). Research suggest that Prolific provides good quality data (Douglas et al., 2023). Participants accessed the survey via Prolific, while the survey itself was conducted using the Unipark online tool (Tivian XI GmbH, EFS-Survey). When accessing the study via Prolific, a short text at both time points provided information about the study's procedure, inclusion and exclusion criteria, voluntary participation, anonymity, data protection, and compensation for participation. Participants were required to provide their consent before proceeding to complete the questionnaires. Ethical approval was given by the University of Münster. The survey took participants an average of 19.38 min (SD = 13.46) to complete at the first assessment, and 23.45 min (SD = 11.93) at the second assessment. This included two attention check questions, and the questionnaires described below.

2.3. Sample size justification

Sample size considerations were influenced by three factors: (1) recruitment challenges, (2) financial constraints, and (3) heuristic considerations (Lakens, 2022). We preregistered a minimum of 300 participants to ensure stable parameter estimates for the follow-up (Schönbrodt & Perugini, 2013; Wolf et al., 2013), as correlations tend to stabilize at this sample size. This target sample size also aligned with our budget, which allowed for recruiting around 300 participants at two time points given Prolific's payment requirements. We conducted post-hoc power analyses for the longitudinal serial mediation models (see below) using the Monte Carlo Power Analysis for indirect effects (Schoemann et al., 2017). The analysis was based on the sample size of the main analysis (N = 226), the correlations between the constructs in the longitudinal serial mediation models (see Table 2), and their standard deviations (see Table 1). We used 5,000 replications, 20,000 Monte Carlo draws per replication, and 99% confidence intervals (α = .01). A separate power analysis was performed for each longitudinal serial mediation model. In none of the models, did the power for the indirect effects of comparison discrepancy and affective comparison impact (A1 × B1; A2 × B2) meet the threshold for sufficient power, and the power for the serial indirect effects (A1 ×

Table 1. Sociodemographic variables and descriptive statistics for all constructs for both time points.

Demographic characteristic	T1 ($N = 315$)		T2 ($N = 226$)	
	M	*SD*	*M*	*SD*
Age	33.16y (18–77y)	10.88	34.52 (18–74y)	11.09
	n	%	*n*	%
Female gender	136	43.17	108	47.79
Male gender	178	56.51	118	52.21
Other gender	1	0.32	0	0
Education				
Less than high school	3	0.95	1	0.44
High school degree or equivalent	53	16.83	44	19.47
Some college but no degree	37	11.75	20	8.85
Associate degree	9	2.86	11	4.87
Bachelor´s degree	159	50.48	115	50.88
Graduate degree	54	17.14	35	15.49
Marital Status				
Married / domestic partnership	143	45.40	106	46.90
Widowed	5	1.59	6	2.65
Divorced	6	1.90	4	1.77
Separated	7	2.22	4	1.77
Single / never married	154	48.89	106	46.90
Household				
Living alone	44	13.97	31	13.72
Living with others	271	86.03	195	86.28
Currently in psychotherapy				
Yes	31	9.84	22	9.73
No	284	90.16	204	90.27
Traumatic event				
Yes	277	87.94	195	86.28
No	38	12.06	31	13.72
Time since traumatic events				
< 6 months	133	48.01	33	16.92
6–12 months	30	10.83	51	26.15
1–5 years	46	16.61	45	23.08
5–10 years	27	9.75	26	13.33
10–20 years	29	10.47	22	11.28
> 20 years	12	4.33	18	9.23
Trauma category				
Accident	21	7.60	11	5.64
Illness	8	2.89	7	3.60
Loss child	5	1.81	7	3.60
Loss friend	15	5.42	10	5.13
Loss parent	42	15.16	25	12.82
Loss relative	50	18.10	22	11.28
Loss sibling	17	6.14	13	6.67
Loss significant other	18	6.50	20	10.26
Physical abuse	7	2.53	7	3.60
Rejection/separation-related	13	4.70	15	7.69
Sexual abuse	9	3.25	7	3.60
Verbal abuse	4	1.44	4	2.10
Violence	7	2.53	6	3.10
Witnessing a traumatic event	5	1.81	4	2.10
Other	56	20.22	37	18.97
	M	*SD*	*M*	*SD*
Comparison frequency	2.51	1.03	2.24	1.44
Comparison discrepancy	2.57	0.96	2.36	1.19
Comparison affective impact	−0.56	0.81	−0.48	0.83
Prolonged grief	11.77	5.36	7.27	5.80
PTSD	11.24	6.99	8.83	7.07
Depression	2.94	1.83	2.47	1.83
Anxiety	2.96	1.78	2.34	1.85
Somatic Symptoms	11.32	7.10	9.70	7.09
Loneliness	5.71	1.97	5.56	2.13
Worry	16.42	5.36	16.59	6.13
Rumination	7.06	3.62	6.53	3.82
Avoidance	54.57	11.77	52.46	13.49
Childhood adversity	6.98	4.47	–	–

A3 × B2) was always below 50%. Therefore, we acknowledge that our sample was underpowered for the mediation models, but should be sufficiently powered for the other analyses.

2.4. Participants

Inclusion criteria required participants to be at least 18 years old, have sufficient English proficiency, and the affirmative answer to the question of whether they had lost a close other within the last three months. $N = 332$ participants met these criteria, passed at least one of two attention checks, and were compensated with £3.50 (GBP). Further exclusion criteria were based on time limits for survey completion, with surveys excluded if completed in less than 8 min ($n = 10$) or more than 1 h ($n = 5$). Additionally, as preregistered, we examined our data for careless/inconsistent effort in responding to our online survey (Curran, 2016; Ward & Meade, 2023) using the *Careless* package in R (Yentes & Wilhelm, 2021). Data from participants with careless response patterns showing no variability ($n = 2$) were excluded, leaving a final dataset of $N = 315$ participants. Six months later, $N = 235$ of these participants completed the follow-up survey, passed attention checks, and were compensated with £4.00 (GBP). Exclusions at this stage included duplicate datasets ($n = 2$), responses completed too quickly ($n = 3$) or too slowly ($n = 2$), and careless responding ($n = 2$), resulting in a final dataset of $N = 226$ participants. Figure 2 shows the flow-chart of participant inclusion into the current study.

2.5. Demographics and retention rate

Table 1 provides demographic information for participants at both time points (T1: $N = 315$; T2: $N = 226$), with a longitudinal retention rate of 71.75%. The sample was predominantly young (mean age of 33.16 years at T1), with a nearly equal gender distribution between females and males, and the majority holding at least a bachelor's degree indicating high educational status of most participants. Most participants were either single or in a marriage/domestic partnership, with a small proportion living alone. About 10% of participants were currently in psychotherapy at both time points. Dropout analyses indicated that participants who did not complete the follow-up survey were significantly more likely to be male ($p = .007$), but no other demographic or mental health differences were observed (see Table 1 for demographic variables). Only data from the $N = 226$ participants who completed both time points were used for the main analyses.

3. Materials

The Comparison Standards Scale for Well-being (CSS-W) measures the frequency, discrepancy, and affective impact of upward and downward social, temporal, counterfactual, and expectation-based comparisons concerning an individual's well-being (Morina &

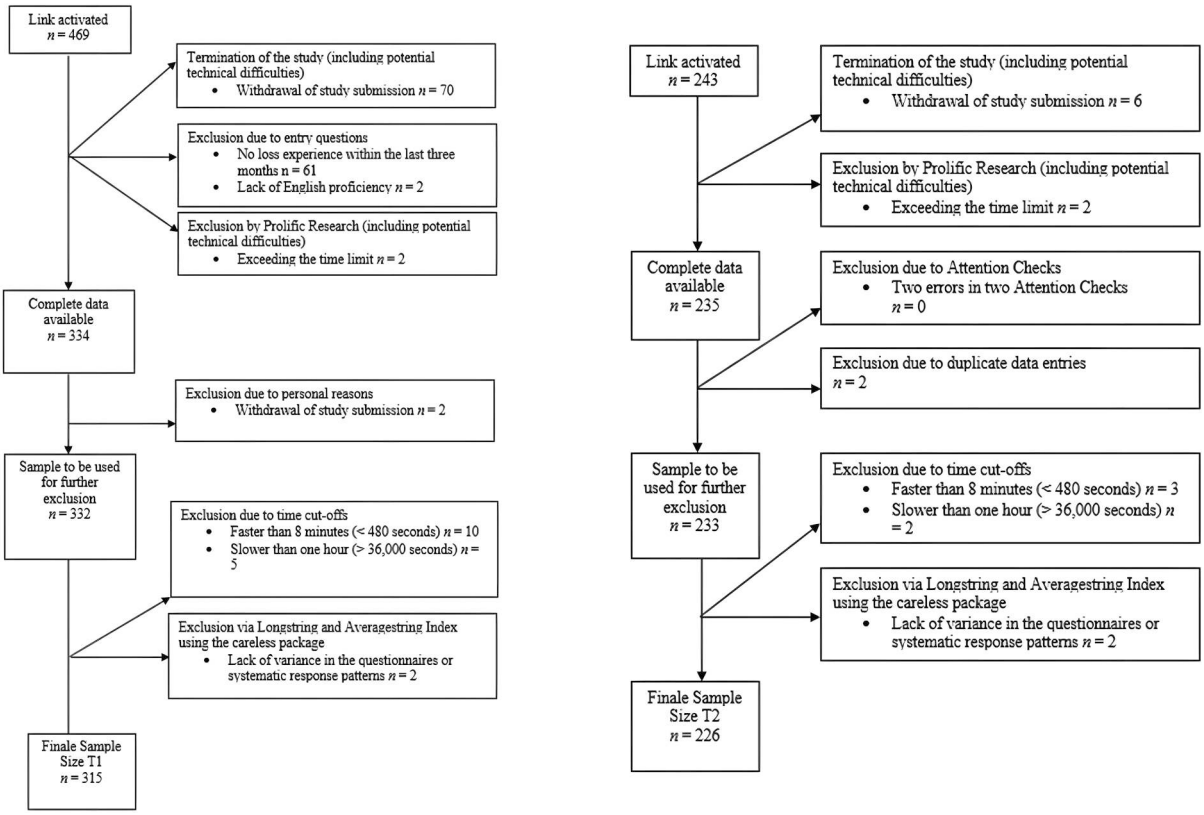


Figure 2. Participant recruitment and retention Flowchart for T1 (left panel) and T2 (right panel).

Schlechter, 2023). The CSS-W comprises three parts: (1) 14 obligatory items that assess the frequency of well-being comparisons in the past three weeks on a six-point Likert scale (0 = 'not at all' to 5 = 'very often'); (2) 14 sub-items that assess the discrepancy from the standard on a six-point Likert scale (0 = 'not at all' to 5 = 'much better' *or* 'much worse'), applicable when the frequency question is answered with a score above 0; and (3) 14 potential sub-items that tap into the affective impact on a bipolar seven-point Likert scale for affective impact (−3 = 'much worse' to +3 = 'much better'), applicable when the frequency question is answered with a score above 0. The structure of the CSS-W is two-dimensional and captures the two correlated latent factors aversive and appetitive comparisons. Upward social, past temporal, counterfactual, and expectation-based comparisons, as well as downward prospective temporal comparisons constitute the aversive comparison factor as these standards threaten the comparers motives (Morina, 2021). Consistent with our research aims, we applied only the aversive well-being comparison subscale. We found acceptable internal consistencies at both time points for comparison frequency (α_{t1} = .77, α_{t2} = .84), discrepancy (α_{t1} = .70, α_{t2} = .80), and affective impact (α_{t1} = .78, α_{t2} = .82).

3.1. Mental health outcome variables

The International Grief Questionnaire (IGQ) assesses symptoms of prolonged grief disorder according to the ICD-11 (Hyland et al., 2024). It consists of 5 items that measure various aspects of prolonged grief disorder, including yearning for the deceased and difficulty accepting the reality of the loss. Items are scored on a 5-point Likert scale, ranging from 1 = 'not at all' to 5 = 'extremely'. Additional dichotomous items assess the level of impairment following the loss. A probable prolonged grief disorder is determined either by a strict assessment algorithm, requiring at least one core symptom item and one additional symptom item of the subscales to be rated ≥ 3 ('often') (n_{t2} = 61; 27.0%), or by a moderate algorithm, where a rating of ≥ 2 ('sometimes') suffices (n_{t2} = 99; 43.8%). For further analyses in this study total scores were used, for which internal reliabilities at both time points were good (α_{t1} = .88, α_{t2} = .93).

The International Trauma Questionnaire (ITQ) was used to assess PTSD symptoms (Cloitre et al., 2018). It starts with identifying the respondent's index traumatic event, which could be a different event than the loss (see Table 1 for trauma categories). Participants were then asked to respond to all questions in reference to this event. The ITQ comprises six items that evaluate PTSD symptoms according to the International Classification of Diseases-11 (World Health Organization, 2018). Items are

evaluated based on the level of distress they have caused in the last month on a five-point scale (0 = 'not at all' to 4 = 'extremely'). Out of the $n = 277$ individuals who reported a potentially traumatic event at T1, $n = 157$ (56.7%) met the criteria for probable PTSD based on questionnaire cut-offs of the subscales (a score above 2 on at least one item in each symptom category: reexperiencing, avoidance, and sense of current threat). At T2, of the 195 individuals who reported a potentially traumatic event, $n = 108$ (55.4%) met the criteria for probable PTSD. Importantly, the high prevalence estimates may result from the absence of an assessment of functional impairment, as well as the possibility that individuals with more severe symptoms were more likely to participate. For further analyses in this study, we used ITQ total scores. Internal consistency in the present study was good at both time points ($\alpha_{t1} = .87$, $\alpha_{t2} = .88$).

The Patient Health Questionnaire-4 (PHQ-4) was used to assess depression and anxiety symptoms (Löwe et al., 2010). This four-item questionnaire assesses symptoms on a four-point scale (0 = 'not at all' to 3 = 'nearly every day'). Symptom endorsement of the last two weeks is assessed, concerning the core symptoms of depression and anxiety (loss of interest, depressed mood, anxiety, constant worries). Sum scores of ≥ 2 are cut-off values distinguishing the normal range from probable cases of depression (exceeded by $n_{t1} = 171$; 54.3%, $n_{t2} = 105$; 46.5%) as well as anxiety (exceeded by $n_{t1} = 113$; 35.9%, $n_{t2} = 81$; 35.8%). For further analyses in this study, total scores were used. Internal consistency was good for the depression subscale (PHQ-2; $\alpha_{t1} = .87$, $\alpha_{t2} = .88$) and the anxiety subscale (GAD-2, $\alpha_{t1} = .82$, $\alpha_{t2} = .84$).

The Somatic Symptom Scale-8 (SSS-8; Gierk et al., 2014) and was used to assess somatic symptoms for the last four weeks on a three-point scale (0 = 'not bothered at all' to 4 = 'bothered a lot'). Scores above 16 are considered clinically high values (exceeded by $n_{t1} = 54$; 17.1%, $n_{t2} = 49$; 15.6%). Total scores were used for subsequent analyses. Internal consistency in the present study was good at both time points ($\alpha_{t1} = .87$, $\alpha_{t2} = .85$).

The UCLA Loneliness Scale-3 (UCLA-3; Russell, 1996) was employed to evaluate loneliness on a global level. It comprises 3 items with a four-point scale (1 = 'never' to 4 = 'often'). Total scores were used for all analyses. Internal consistency in the present study was good at both time points ($\alpha_{t1} = .86$, $\alpha_{t2} = .89$).

3.2. Transdiagnostic predictor variables

The short form of the Penn State Worry Questionnaire (PSWQ) was used to measure excessive worry (Meyer et al., 1990; Topper et al., 2014). It consists of 5 items assessing the tendency, intensity and uncontrollability of worry on a 5-point scale (1 = 'not at all typical' to 5 = '*very* typical'). Internal consistency in the present study was good at both time points ($\alpha_{t1} = .91$, $\alpha_{t2} = .94$).

The Brooding subscale of the Response Style Questionnaire (RSQ) was applied to evaluate the tendency to engage in brooding responses to depressed mood (Nolen-Hoeksema, 1991). This reliable five-item scale instructs participants to report how they typically react to depressed mood on a three point-scale (0 = 'never' to 3 = 'always'). Internal consistency in the present study was good at both time points ($\alpha_{t1} = .83$, $\alpha_{t2} = .85$).

The Multidimensional Experiential Avoidance Questionnaire (MEAQ; Ottenbreit & Dobson, 2004) is a multidimensional measure of avoidance, comprises 15 items on a 6-point scale (1 = 'strongly disagree' to 6 = 'strongly agree'). This scale comprises six avoidance domains: behavioural avoidance, distress aversion, procrastination, distraction and suppression, repression and denial, as well as distress endurance. Internal consistency in the present study was good at both time points ($\alpha_{t1} = .86$, $\alpha_{t2} = .88$).

The Comprehensive Adversity Measure (CAM) comprises 22 dichotomous yes/no items assessing childhood adversity (Schlechter, Lutz, et al., 2024). These items inquire whether a range of adverse events such as sexual abuse or interpersonal violence were experienced before the age of 18. A sum score of all experienced events formed the childhood adversity indicator. Childhood adversity was only assessed at the first timepoint, as it took place before the age of 18. Internal consistency was good, $\alpha = .81$.

3.3. Analysis procedure

Data analyses were conducted using R (R Core Team, 2025). Descriptive statistics were calculated for both time points (T1 and T2). Paired-sample t-tests assessed significant differences between T1 and T2. Pearson correlation coefficients were calculated to examine associations between comparison process variables (e.g. comparison frequency, discrepancy, and affective impact) and mental health outcomes (e.g. grief, PTSD, depression, anxiety, somatic symptoms, and loneliness) at both time points. Four types of regression models were used to examine predictive relationships and the unique role of comparative thinking beyond worry, rumination, avoidance, and childhood adversity: (1) T1 variables predicting T2 outcomes (longitudinal associations), (2) T1 variables predicting T2 outcomes with autoregressive effects to account for stability over time, (3) T2 variables predicting other T2 variables (concurrent associations, as comparative thinking was assessed for the past three weeks), and (4) models with T2 variables predicting other T2 variables including autoregressive pathways from T1. Predictors in all models included

aversive comparisons, worry, rumination, avoidance, and childhood adversity, with two models adjusted for the autoregressive effects of our outcome variables (i.e. grief, PTSD, depression, anxiety, somatic symptoms, & loneliness). Models without autoregressive pathways were estimated to examine whether predictor variables were associated with subsequent levels of psychopathology over time. Autoregressive models were estimated to assess whether predictor variables explained variance in mental health outcomes beyond baseline symptom levels, thereby accounting for symptom stability across time. Examining both model types allows for the distinction between longitudinal associations with later symptom severity and associations with changes in symptoms over time. Regression models were conducted separately for each outcome measure. Mediation analyses using the *lavaan* package (Rosseel, 2012) tested indirect effects cross-sectionally (at T2) and longitudinally (T1 to T2), examining whether relationships between comparison frequency (X) and mental health outcomes (Y) were mediated by comparison discrepancy (M1) and affective impact (M2), with indirect effects estimated via 10,000 bootstrap samples. Cross-sectional mediation models were estimated to examine associations among variables measured within the same assessment window. This is relevant because the comparison variables referred to experiences during the preceding three weeks, making concurrent associations informative for that same timeframe. Results were primarily evaluated at our preregistered alpha level of 0.01. However, *p*-values at $\alpha = 0.05$ are also reported and interpreted with caution, given our post-hoc power evaluation indicating potential underpowering of our study.

4. Results

4.1. Descriptive statistics

Descriptive statistics of all constructs can be found in Table 1. T-tests revealed that comparison frequency slightly decreased across time ($t = 2.47$, df = 443.39, $p < .05$), as did comparison discrepancy ($t = 2.28$, df = 440.63, $p < .05$) and comparison affective impact ($t = -2.04$, df = 449.55, $p < .05$), but not at our preregistered alpha level ($p = .01$). Grief symptoms decreased significantly ($t = 8.22$, df = 449.08, $p < .001$), as did symptoms of PTSD ($t = 3.51$, df = 449.99, $p < .001$), depression ($t = 2.77$, df = 449.78, $p < .01$), and anxiety ($t = 2.98$, df = 449.74, $p < .01$). Loneliness scores and somatic symptoms did not decrease over time *p*s > .084.

4.2. Correlations among variables

Table 2 presents the correlations between the comparison variables and mental health outcomes. Our

Table 2. Correlations among comparison variables and mental health variables.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Freq_t1																	
2. Disc_t1	.89*																
3. Affect_t1	−.46*	−.46*															
4. IGQ_t1	.38*	.34*	−.20*														
5. ITQ_t1	.44*	.40*	−.29*	.63*													
6. PHQ_t1	.40*	.40*	−.42*	.48*	.56*												
7. GAD_t1	.48*	.42*	−.40*	.41*	.55*	.66*											
8. SSS-8_t1	.51*	.44*	−.40*	.49*	.56*	.57*	.62*										
9. UCLA3_t1	.42*	.40*	−.36*	.29*	.30*	.52*	.36*	.44*									
10. Freq_t2	.59*	.53*	−.27*	.39*	.45*	.42*	.44*	.45*	.32*								
11. Disc_t2	.59*	.57*	−.31*	.38*	.45*	.42*	.41*	.47*	.32*	.93*							
12. Affect_t2	−.32*	−.26*	.51*	−.16*	−.23*	−.30*	−.29*	−.29*	−.25*	−.37*	−.40*						
13. IGQ_t2	.30*	.24*	−.10	.53*	.53*	.34*	.38*	.46*	.12	.45*	.45*	−.18*					
14. ITQ_t2	.41*	.36*	−.19*	.53*	.64*	.47*	.49*	.50*	.28*	.50*	.48*	−.30*	.61*				
15. PHQ_t2	.45*	.42*	−.29*	.37*	.46*	.60*	.48*	.53*	.45*	.59*	.58*	−.42*	.49*	.57*			
16. GAD_t2	.48*	.40*	−.30*	.45*	.51*	.47*	.62*	.54*	.34*	.55*	.52*	−.43*	.58*	.65*	.72*		
17. SSS-8_t2	.50*	.42*	−.34*	.42*	.50*	.52*	.56*	.78*	.47*	.56*	.56*	−.42*	.49*	.61*	.72*	.66*	
18.UCLA3_t2	.45*	.39*	−.26*	.27*	.34*	.44*	.37*	.43*	.69*	.47*	.46*	−.35*	.30*	.39*	.59*	.47*	.54*

Note. * indicates $p < .01$. Freq = Comparison Frequency; Disc = Comparison discrepancy; Affect = Comparison affective impact; IGQ = The International Grief Questionnaire (Grief); ITQ = International Trauma Questionnaire; PHQ = Patient Health Questionnaire-2 (Depression); GAD = Generalized Anxiety Disorder 2-item (Anxiety); SSS −8 = Somatic Symptom Scale-8; ULCA = UCLA Loneliness Scale-3; t1 = timepoint 1; t = timepoint 2.

constructs showed correlations in the expected direction at our preregistered alpha level of .01. Specifically, comparison frequency had a strong positive correlation with comparison discrepancy and a moderate negative correlation with comparison affective impact, both cross-sectionally and longitudinally. Additionally, it was positively correlated with grief reactions, PTSD, depression, anxiety, somatic symptoms, and loneliness at both time points. Comparison discrepancy was positively associated with grief reactions, PTSD, depression, anxiety, somatic symptoms, and loneliness, and negatively correlated with comparison affective impact, both cross-sectionally and longitudinally. Comparison affective impact showed negative correlations with grief reactions, PTSD, depression, anxiety, somatic symptoms, and loneliness, both cross-sectionally and longitudinally. Moreover, grief symptoms were positively related to PTSD, depression, anxiety, somatic symptoms, and loneliness at both time points.

4.3. Regression models

Table 3 shows the results of the regression models accounting for the other predictor variables. In the regression models, we did not include comparison discrepancy given its high overlap with comparison frequency. Apart from this, multicollinearity was not detected in any model, with all variance inflation factors being below 3.

4.3.1. T1 variables predicting T2 variables (Regression models 1)

Comparison frequency was significantly positively related to anxiety, somatic symptoms, and loneliness. Worry showed significant positive relationships with PTSD, depression, and anxiety. Rumination exhibited highly significant positive relationships with grief reactions, PTSD, depression, anxiety, somatic symptoms, and loneliness. Avoidance had a significant positive relationship with depression, but showed no significant effects on other outcomes. Comparison affective impact was significantly positively related to grief reactions and PTSD symptoms (however, the directionality was contrary to single associations and may represent a statistical artifact, although multicollinearity was not detected). Childhood adversity demonstrated significant positive relationships with PTSD, depression, anxiety, somatic symptoms, and loneliness.

4.3.2. T1 variables predicting T2 variables with autoregressive effects (Regression models 2)

Autoregressive effects were significant for all outcomes. Comparison frequency was significantly positively related to depression, anxiety, and loneliness. Worry showed a significant positive relationship with anxiety. Rumination was significantly positively related to PTSD, depression, and anxiety. Avoidance showed no significant relationship with any of the outcomes. Comparison affective impact had a significant positive relationship with PTSD (however, the directionality was contrary to single associations and may represent a statistical artifact, and was only significant at $p < .05$). Affective impact showed no significant relationships with the other outcomes. Childhood adversity showed no significant relationships with the outcomes.

Table 3. Regression models with all predictors with and without autoregressive pathways.

Model	Comparison frequency	Worry	Rumination	Avoidance	Comparison affective impact	Childhood adversity	Autoregressive
T1T2	*Estimate (SE), p*						
Grief	0.50 (0.42), $p = .23$	0.13 (0.09), $p = .14$	0.38 (0.14), $p < .01$	0.05 (0.04), $p = .12$	1.07 (0.49), $p < .05$	0.15 (0.09), $p = .1$	–
PTSD	0.86 (0.47), $p = .07$	0.20 (0.1), $p = .05$	0.61 (0.15), $p < .001$	0.02 (0.04), $p = .62$	1.05 (0.55), $p = .06$	0.32 (0.1), $p < .01$	–
Depression	0.22 (0.12), $p = .07$	0.05 (0.02), $p < .05$	0.14 (0.04), $p < .001$	0.02 (0.01), $p < .05$	0.03 (0.14), $p = .85$	0.06 (0.02), $p < .05$	–
Anxiety	0.27 (0.11), $p < .05$	0.1 (0.02), $p < .001$	0.12 (0.04), $p < .01$	0.00 (0.01), $p = .99$	0.05 (0.13), $p = .70$	0.07 (0.02), $p < .01$	–
Somatic	1.19 (0.44), $p < .01$	0.12 (0.09), $p = .20$	0.5 (0.14), $p < .001$	0.05 (0.04), $p = .14$	−0.21 (0.51), $p = .68$	0.38 (0.09), $p < .001$	–
Loneliness	0.38 (0.14), $p < .01$	0.01 (0.03), $p = .76$	0.17 (0.05), $p < .001$	0.02 (0.01), $p = .17$	0.07 (0.17), $p = .68$	0.09 (0.03), $p < .01$	–
T1T2							
Grief	0.28 (0.39), $p = .48$	0.12 (0.08), $p = .17$	0.19 (0.13), $p = .15$	0.02 (0.03), $p = .62$	0.88 (0.46), $p = .06$	0.09 (0.08), $p = .26$	0.41 (0.07), $p < .001$
PTSD	0.52 (0.43), $p = .22$	0.13 (0.09), $p = .15$	0.32 (0.14), $p < .05$	0.02 (0.04), $p = .57$	1.01 (0.5), $p < .05$	0.16 (0.09), $p = .08$	0.44 (0.06), $p < .001$
Depression	0.24 (0.11), $p < .05$	0.03 (0.02), $p = .18$	0.08 (0.04), $p < .05$	0.01 (0.01), $p = .12$	0.16 (0.13), $p = .22$	0.03 (0.02), $p = .2$	0.34 (0.07), $p < .001$
Anxiety	0.22 (0.11), $p < .05$	0.07 (0.02), $p < .01$	0.08 (0.04), $p < .05$	0 (0.01), $p = .95$	0.12 (0.13), $p = .35$	0.04 (0.02), $p = .08$	0.33 (0.07), $p < .001$
Somatic	0.62 (0.36), $p = .08$	0.11 (0.08), $p = .17$	0.07 (0.12), $p = .56$	0.03 (0.03), $p = .31$	0.33 (0.42), $p = .43$	0.11 (0.08), $p = .16$	0.64 (0.06), $p < .001$
Loneliness	0.27 (0.12), $p < .05$	0.02 (0.03), $p = .43$	0.07 (0.04), $p = .08$	0.01 (0.01), $p = .54$	0.24 (0.14), $p = .09$	0.03 (0.03), $p = .20$	0.58 (0.06), $p < .001$
T2T2							
Grief	0.96 (0.36), $p < .01$	0.16 (0.08), $p < .05$	0.3 (0.13), $p < .05$	0.06 (0.03), $p < .05$	0.7 (0.44), $p = .12$	0.04 (0.09), $p = .65$	–
PTSD	0.75 (0.4), $p = .06$	0.14 (0.08), $p = .10$	0.51 (0.15), $p < .001$	0.11 (0.03), $p < .01$	0.1 (0.49), $p = .84$	0.2 (0.09), $p < .05$	–
Depression	0.41 (0.1), $p < .001$	0.05 (0.02), $p < .05$	0.11 (0.04), $p < .01$	0.01 (0.01), $p = .38$	−0.27 (0.12), $p < .05$	0.03 (0.02), $p = .19$	–
Anxiety	0.25 (0.1), $p < .01$	0.1 (0.02), $p < .001$	0.07 (0.04), $p = .06$	0.01 (0.01), $p = .23$	−0.32 (0.12), $p < .01$	0.05 (0.02), $p < .05$	–
Somatic	1.04 (0.37), $p < .01$	0.16 (0.08), $p < .05$	0.57 (0.13), $p < .001$	0.02 (0.03), $p = .57$	−0.98 (0.45), $p < .05$	0.29 (0.09), $p < .001$	–
Loneliness	0.21 (0.12), $p = .09$	0.01 (0.03), $p = .64$	0.22 (0.05), $p < .001$	0.01 (0.01), $p = .28$	−0.16 (0.15), $p = .28$	0.06 (0.03), $p < .05$	–
T2T2							
Grief	0.72 (0.34), $p < .05$	0.13 (0.07), $p = .06$	0.22 (0.12), $p = .08$	0.04 (0.03), $p = .22$	0.52 (0.41), $p = .21$	−0.02 (0.08), $p = .79$	0.36 (0.06), $p < .001$
PTSD	0.35 (0.36), $p = .33$	0.08 (0.08), $p = .30$	0.39 (0.13), $p < .01$	0.09 (0.03), $p < .01$	−0.01 (0.44), $p = .98$	0.05 (0.09), $p = .54$	0.4 (0.06), $p < .001$
Depression	0.37 (0.09), $p < .001$	0.03 (0.02), $p = .12$	0.09 (0.03), $p < .01$	0 (0.01), $p = .57$	−0.23 (0.11), $p < .05$	0 (0.02), $p = .96$	0.31 (0.05), $p < .001$
Anxiety	0.23 (0.09), $p < .05$	0.06 (0.02), $p < .01$	0.05 (0.03), $p = .17$	0.01 (0.01), $p = .29$	−0.31 (0.11), $p < .01$	0.02 (0.02), $p = .35$	0.31 (0.06), $p < .001$
Somatic	0.69 (0.29), $p < .05$	0.05 (0.06), $p = .39$	0.33 (0.11), $p < .01$	0.02 (0.03), $p = .47$	−0.91 (0.35), $p < .05$	0.02 (0.07), $p = .82$	0.56 (0.05), $p < .001$
Loneliness	0.19 (0.1), $p = .06$	0.01 (0.02), $p = .58$	0.15 (0.04), $p < .001$	0.01 (0.01), $p = .45$	−0.08 (0.12), $p = .51$	−0.01 (0.02), $p = .72$	0.55 (0.05), $p < .001$

Note. Each column displays the regression estimate (standard error), and *p*-value.

4.3.3. T2 variables predicting T2 variables (Regression models 3)

Comparison frequency was significantly positively related to grief reactions, depression, anxiety, and somatic symptoms. Worry showed significant positive relationships with grief reactions, depression, anxiety, and somatic symptoms. Rumination had significant positive relationships with grief, PTSD, depression, somatic symptoms, and loneliness. Avoidance was significantly positively related to grief reactions and PTSD symptoms. Comparison affective impact showed significant negative relationships with depression, anxiety, and somatic symptoms. Childhood adversity had significant positive relationships with PTSD, anxiety, somatic symptoms, and loneliness.

4.3.4. T2 variables predicting T2 variables with autoregressive effects (Regression models 4)

Autoregressive effects were significant for all outcomes. Comparison frequency was significantly positively related to grief reactions, depression, anxiety, and somatic symptoms. Worry showed a significant positive relationship with anxiety. Rumination had significant positive relationships with PTSD, depression, somatic symptoms, and loneliness. Avoidance showed no significant relationships with any of the outcomes, apart from a positive association with PTSD. Comparison affective impact was significantly negatively related to depression, anxiety, and somatic symptoms. Childhood adversity showed no significant relationships with any of the outcomes.

4.4. Mediation models

Both cross-sectionally and longitudinally (T1 to T2), we examined whether relationships between comparison frequency (X) and mental health outcomes (Y) were mediated by comparison discrepancy (M1) and affective impact (M2). None of the outcome variables showed a significant effect in the longitudinal mediation models. However, in the cross-sectional mediation models using only T2 variables, we found significant indirect effects for depression (indirect effect = 0.13, CI: [−0.30, 0.30], $p = .036$), anxiety (indirect effect = 0.16, CI: [−0.30, 0.35], $p = .030$), and somatic symptoms (indirect effect = 0.54, CI: [−0.12, 1.20], $p = .036$). These effects were mediated through the pathways involving comparison frequency, discrepancy, and affective impact. However, the serial mediation effects reached significance at $\alpha = .05$, but not at our preregistered threshold of $\alpha = .01$. The paths leading to depression as the outcome are visually depicted in Figure 3 for the cross-sectional model at T2, and they were similar for anxiety and somatic symptoms. In most other models, comparison frequency and affective impact predicted the outcomes, whereas comparison discrepancy did not.

5. Discussion

We examined the role of well-being comparisons in mental health symptoms at two time points over a six-month period in recently bereaved individuals. Our findings reveal the potential impact of comparison processes on mental health outcomes in this vulnerable population.

5.1. General findings

Regarding mental health symptoms, grief reactions, PTSD, depression, and anxiety exhibited significant reductions over time. This aligns with prior research indicating that a majority of individuals show resilient trajectories following the loss of a significant other (Bonanno & Malgaroli, 2020; Djelantik et al., 2022). Likewise, over the six-month period, comparison frequency, comparison discrepancy, and comparison affective impact all decreased slightly. This aligns with expectations of gradual emotional adjustment following loss, wherein bereaved individuals may engage less in comparative thinking as they progress

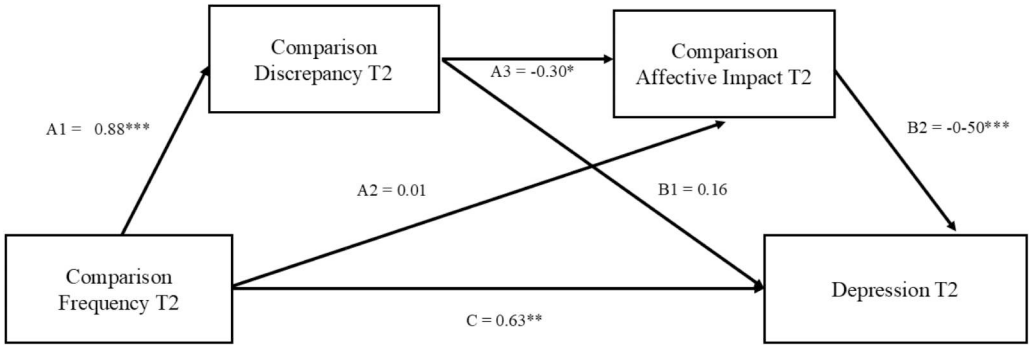


Figure 3. Serial multiple mediation models of the association between aversive comparison frequency at t2 (X) and depression symptoms and at t2 (Y). Comparison discrepancy serves as fist mediator (M1) and comparison affective impact as second mediator (M2). Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

through their grieving process. However, while these variables showed a decrease, all comparison process variables (i.e. comparison frequency, discrepancy, and affective impact) were consistently correlated with all mental health outcomes (i.e. prolonged grief reactions, PTSD, depression, anxiety, somatic symptoms, and loneliness) at both time points. Most effect sizes were moderate, with some reaching a large magnitude (e.g. $r = .59$ between T2 comparison frequency and depression). While aversive comparisons may decrease over time, their continued association with ongoing distress indicates they may contribute to the persistence of grief reactions and associated symptoms.

5.2. Incremental contribution of aversive well-being comparisons

Our results revealed a nuanced role of comparative thinking in mental health outcomes following bereavement. This role remained evident even when controlling for worry, rumination, avoidance, and childhood adversity. Comparison frequency consistently emerged as an important predictor of distress across models, particularly in relation to anxiety, depression, somatic symptoms, and loneliness both cross-sectionally and longitudinally. These symptoms encompass cognitive and emotional patterns and may, therefore, be influenced by the repetitive and evaluative nature of comparison frequency. Comparison affective impact also emerged as predictor in correlational analyses. In addition, in regression models (Models 3 & 4) affective impact at T2 was related to depression, anxiety, and somatic symptoms at T2. Affective impact directly reflects the psychological consequences of frequent comparisons (Morina, 2021). Comparison affective impact may have immediate and cumulative effects on mental health, as it informs how individuals perceive their current well-being and shapes subsequent emotional and behavioural responses. Our findings are especially significant because comparative thinking accounted for unique variance to mental health outcomes beyond established risk factors such as worry, rumination, avoidance, and childhood adversity (McEvoy et al., 2013; McLaughlin et al., 2012; Morina, 2011). Importantly, associations remained significant when we accounted for autoregressive effects. This underscores the important role of aversive well-being comparisons in shaping distress following bereavement.

5.3. Comparative thinking and rumination

In models with all predictors, comparison frequency over the last three weeks as assessed at T1 did not significantly predict grief reactions longitudinally, whereas rumination emerged as the most important factor across models. However, comparative frequency at T2 emerged as the only significant variable when accounting for autoregressive effects. This may be attributable to the time frame. The retrospective window at T1 may not capture the cumulative effects of comparative thinking over time, as the bereavement process is still in its early stages, and comparative thinking in the last three weeks may not be predictive of grief symptoms six months later. Rumination was assessed at a more trait-like, habitual level and may, thus, have exerted longer-term effects. Yet, in the concurrent models at T2, specific comparisons in the last three weeks may become more representative of the individual's cognitive processes. This was also reflected in findings concerning comparison affective impact. The finding that it was negatively associated with depression, anxiety, and somatic symptoms only concurrently at T2 suggests that the immediate affective impact of well-being comparisons may have a more direct and short-term influence on mental health than a lasting longitudinal effect that cumulates over six months (when regressed with more trait like predictors). While rumination may also amplify well-being comparisons or vice versa (Schlechter & Morina, 2023), as evidenced by their bidirectional associations, they remain distinct constructs with unique implications for grief. Brooding rumination, sometimes referred to as 'moody pondering' (Treynor, 2003), involves passively focusing on distress and its causes or consequences without necessarily engaging in explicit comparative thinking (Nolen-Hoeksema et al., 2008). In the context of grief symptoms, brooding rumination may manifest as repetitive cognitions about the loss, its causes, or self-directed blame. In contrast, well-being comparisons involve evaluating (dis-)similarities between one's current well-being and a standard, such as pre-loss functioning or the perceived well-being of others (Morina, 2021). While brooding rumination may sometimes include comparative elements (e.g. 'Why am I struggling more than others?'; Nolen-Hoeksema, 1991), well-being comparisons are explicitly focused on cognitive evaluations of potential well-being discrepancies. Brooding rumination perpetuates negative affect by focusing on the loss, self-blame, or unresolved questions, potentially exacerbating grief reactions and related symptoms. In contrast, aversive well-being comparisons may influence grief by accentuating discrepancies between one's current state and desired states, which can amplify feelings of inadequacy and distress and negatively influence coping efforts. Importantly, these processes might interact, with aversive comparisons leading to negative mood and rumination. Conversely, rumination

may intensify the emotional impact of aversive comparisons. To assess this interaction, ecological momentary assessment (EMA) is needed to evaluate the daily occurrence of brooding rumination and well-being comparisons, including their potential sequential order. Additionally, experimental research is necessary to further elucidate the mechanisms underlying their influence on grief symptoms.

5.4. Comparative thinking in the context of other predictors

Our findings indicate that comparative thinking is an important factor to consider alongside the other predictors examined in this study. In line with prior work, worry was consistently associated with outcome variables, highlighting its role in maintaining anticipatory fears and catastrophic thoughts about one's capacity to recover from loss (McEvoy et al., 2013). Rumination demonstrated even broader associations, significantly predicting grief, PTSD, depression, anxiety, somatic symptoms, and loneliness. This highlights how repetitive and uncontrollable focus on unresolved loss can maintain and increase distress (Nolen-Hoeksema et al., 2008). While avoidance showed relatively limited effects compared to other variables, its positive association with depression and PTSD at specific time points underscores its role as a maladaptive coping strategy (Eisma & Stroebe, 2017). In addition, avoidance was broadly assessed as experiential avoidance and may thus not accurately capture disorder-specific aspects of avoidance. Childhood adversity emerged as a significant predictor of PTSD, depression, anxiety, and somatic symptoms (McLaughlin et al., 2012). However, its influence diminished or was much smaller in the autoregressive models, suggesting that its effects may be partially mediated by more proximal factors.

The absence of consistent effects of comparative thinking on PTSD symptoms in the regression models, despite significant bivariate associations, suggests that its relationship with PTSD may be better explained through shared variance with other psychological factors, such as rumination or worry. PTSD is characterized by intrusive re-experiencing, avoidance behaviour and hyperarousal, which may conceptually be more strongly related with rumination and worry (Nolen-Hoeksema et al., 2008). These constructs may mediate or overshadow the influence of comparative thinking when considered simultaneously. Importantly, PTSD symptoms in the present study were not directly related to the loss.

5.5. Serial mediation effects

Our longitudinal mediation models did not yield significant effects. However, cross-sectional models at Time 2 showed nearly significant indirect effects for depression, anxiety, and somatic symptoms. This may correspond with our discussion above concerning the timeframe of comparative thinking. Effects of comparison frequency on depression, anxiety, and somatic symptoms were mediated through comparison discrepancy and comparison affective impact. These findings suggest that well-being comparisons may have an indirect influence on mental health outcomes through their cognitive (i.e. assessment of the comparison discrepancy) and emotional (i.e. the immediate affective reaction) impact. This is consistent with previous studies on well-being comparison that repeatedly found this mediation in the context of various mental health symptoms (Morina & Schlechter, 2023; Schlechter & Morina, 2023). However, the serial mediation effects were significant at the $\alpha = .05$ level and did not meet our preregistered threshold of $\alpha = .01$, indicating that these pathways require further empirical validation, particularly given that post-hoc power analysis suggested the study was underpowered to detect these mediation effects.

5.6. Theoretical contribution

Our study contributes to the theoretical understanding of how aversive well-being comparisons may influence mental health outcomes in the context of bereavement. Our considerations were based on gComp (Morina, 2021), which highlights how comparison selection, perceived discrepancies, and affective and behavioural impact shape outcome variables. Specifically, aversive well-being comparisons not only represent cognitive appraisals of well-being but also elicit affective reactions that intensify grief-related symptoms. The dual process model emphasizes the need for individuals to balance two complementary processes when grieving: (1) loss-oriented coping, which focuses on the emotional aspects of the loss, and (2) restoration-oriented coping, which deals with the adaptation to life without the deceased (Stroebe & Schut, 1999). Aversive well-being comparisons may influence both of these coping processes, and disrupt helpful oscillation as it mostly reflects loss-oriented coping processes. When individuals frequently engage in aversive well-being comparisons, it might exacerbate the loss-oriented aspect of grief, as they become preoccupied with what they have lost, hindering their ability to reorient toward life outside of bereavement. Negative comparison affective impact may obstruct the process of restoration-oriented coping. From a cognitive behavioural perspective (Boelen et al., 2006), frequent engagement in aversive comparisons may evoke and maintain depression, anxiety, and grief. Aversive well-being comparisons may thus fuel maladaptive thinking

patterns and reinforce cognitive distortions that the comparer's well-being is permanently diminished.

5.7. Future research

Future research should further disentangle the role of well-being comparison processes at multiple levels of analysis, ranging from broader longitudinal trajectories to fine-grained within-person dynamics. In particular, micro-level approaches (e.g. ecological momentary assessment) are needed to capture the temporal sequencing and short-term fluctuations of comparison frequency, discrepancy, and affective impact in daily life (Schlechter et al., 2025, 2026). Experimental and intervention studies are also required to clarify causal mechanisms and determine whether modifying maladaptive comparison processes can reduce grief-related distress. Additionally, future work should examine how well-being comparisons operate alongside established transdiagnostic processes such as rumination, worry, and avoidance, and whether they represent partially independent or interacting mechanisms across different symptom domains in bereaved individuals.

5.8. Potential clinical implications

Our findings suggest that aversive well-being comparisons contribute to distress in prolonged grief, underscoring their potential importance as a target for therapeutic intervention. Integrating approaches to reduce comparison frequency, reframe comparison discrepancies, or improve coping with the emotional effects of comparisons could enhance the efficacy of existing treatments for PGD. This aligns well with cognitive behaviour therapy (CBT) interventions, which has been found effective in treating PGD (LaPlante et al., 2024) by targeting maladaptive thoughts and emotions. A targeted treatment focus on aversive well-being comparisons may help patients balance loss-oriented and restoration-oriented stressors, potentially promoting better emotional processing and grief adjustment. However, these claims require validation through experimental and clinical research.

5.9. Strengths and limitations

This preregistered study contributes to the literature by examining the role of aversive well-being comparisons in individuals experiencing loss. It expands upon previous research by considering the impact of well-being comparison processes, including its frequency, discrepancy, and affective impact, and demonstrating its unique contribution to mental health outcomes beyond established risk factors like rumination and worry. By investigating these factors alongside other key transdiagnostic risk factors in bereaved individuals, our work provides valuable insights into the complex cognitive processes that may intensify grief and contribute to mental health problems.

However, our study has some limitations. One notable limitation is the dropout rate of 28%. While mental health variables were not associated with drop-outs, unmeasured variables may have contributed to the drop-out in our study and may thus have influenced the results. Our study was limited to two time-points, six months apart. Relevant short-term fluctuations and within-person person processes thus remain masked in our study. Although our models tested mediational pathways, causality cannot be established. Experimental research is needed to address this limitation. Furthermore, we relied on a convenience sample from Prolific, which restricts the extent to which our findings can be generalized. This concerns sample diversity and recruitment. Although Prolific allows access to a wide range of participants, participants were self-selected and recruited online, which may introduce selection biases (e.g. higher digital literacy, greater willingness to participate in psychological research). In addition, the sample may not fully capture the diversity of bereaved populations in terms of cultural background, socioeconomic status, or bereavement context (e.g. cause of death, relationship to the deceased). Additionally, post-hoc power analyses suggested that our study may not have had sufficient statistical power to detect the mediation effects we hypothesized. This requires caution in interpreting the mediation findings and underscores the need for larger samples in future research. Finally, childhood adversity was assessed retrospectively and may be confounded with mood-related recall bias (Dunn et al., 2018).

6. Conclusion

Our findings contribute to the growing literature on cognitive processes involved in bereavement by emphasizing the role of well-being comparisons in mental health following the loss of a significant other. A better understanding of how comparison processes influence mental health during grief may advance our knowledge of prolonged grief and help inform interventions aimed at enhancing well-being among bereaved individuals.

Author contributions

CRediT: **Pascal Schlechter:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft; **Thole H. Hoppen:** Conceptualization, Investigation, Methodology, Writing – review & editing; **George A. Bonanno:** Conceptualization, Investigation, Methodology, Writing – review & editing; **Nexhmedin Morina:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Writing – original draft.

Open scholarship



This article has earned the Center for Open Science badges for Open Data, Open Materials and Preregistered. The data and materials are openly accessible at https://osf.io/4kdb9/?view_only=4081bb06e5b34f43bcef4f9d6305200e.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data and R code used in this study are openly available.

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References

Akbari, M., & Khanipour, H. (2018). The transdiagnostic model of worry: The mediating role of experiential avoidance. *Personality and Individual Differences*, *135*, 166–172.

Albert, S. (1977). Temporal comparison theory. *Psychological Review*, *84*, 485–503. https://doi.org/10.1037/0033-295X.84.6.485

American Psychiatric Association (Hrsg.). (2013). *Diagnostic and statistical manual of mental disorders: DSM-5* (5th ed). American Psychiatric Association.

Bellet, B. W., LeBlanc, N. J., Nizzi, M.-C., Carter, M. L., van der Does, F. H. S., Peters, J., Robinaugh, D. J., & McNally, R. J. (2020). Identity confusion in complicated grief: A closer look. *Journal of Abnormal Psychology*, *129*(4), 397–407. https://doi.org/10.1037/abn0000520

Bernstein, E. E., Heeren, A., & McNally, R. J. (2019). Reexamining trait rumination as a system of repetitive negative thoughts: A network analysis. *Journal of Behavior Therapy and Experimental Psychiatry*, *63*, 21–27.

Boelen, P. A., van den Hout, M. A., & van den Bout, J. (2006). A cognitive-behavioral conceptualization of complicated grief. *Clinical Psychology: Science and Practice*, *13*(2), 109–128. https://doi.org/10.1111/j.1468-2850.2006.00013.x

Bonanno, G. A., & Malgaroli, M. (2020). Trajectories of grief: Comparing symptoms from the DSM-5 and ICD-11 diagnoses. *Depression and Anxiety*, *37*(1), 17–25. https://doi.org/10.1002/da.22902

Borkovec, T. D., Ray, W. J., & Stober, J. (1998). Worry: A cognitive phenomenon intimately linked to affective, physiological, and interpersonal behavioral processes. *Cognitive Therapy and Research*, *22*(6), 561–576.

Chen, M. A., Lewis, M. R., Chirinos, D. A., Murdock, K. W., & Fagundes, C. (2020). Differential psychological reactions to grief: The role of childhood adversity for depression symptoms among bereaved and non-bereaved adults. *Death Studies*, *44*(12), 778–786. https://doi.org/10.1080/07481187.2019.1614107

Cloitre, M., Shevlin, M., Brewin, C. R., Bisson, J. I., Roberts, N. P., Maercker, A., Karatzias, T., & Hyland, P. (2018). The International Trauma Questionnaire: Development of a self-report measure of ICD-11 PTSD and complex PTSD. *Acta Psychiatrica Scandinavica*, *138*(6), 536–546. https://doi.org/10.1111/acps.12956

Comtesse, H., Smid, G. E., Rummel, A.-M., Spreeuwenberg, P., Lundorff, M., & Dückers, M. L. A. (2024). Cross-national analysis of the prevalence of prolonged grief disorder. *Journal of Affective Disorders*, *350*, 359–365. https://doi.org/10.1016/j.jad.2024.01.094

Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, *66*, 4–19. https://doi.org/10.1016/j.jesp.2015.07.006

Djelantik, A. A. A. M. J., Robinaugh, D. J., & Boelen, P. A. (2022). The course of symptoms in the first 27 months following bereavement: A latent trajectory analysis of prolonged grief, posttraumatic stress, and depression. *Psychiatry Research*, *311*, 114472. https://doi.org/10.1016/j.psychres.2022.114472

Douglas, B. D., Ewell, P. J., & Brauer, M. (2023). Data quality in online human-subjects research: Comparisons between MTurk, Prolific, CloudResearch, Qualtrics, and SONA. *PLoS One*, *18*(3), e0279720. https://doi.org/10.1371/journal.pone.0279720

Dunn, E. C., Soare, T. W., Raffeld, M. R., Busso, D. S., Crawford, K. M., Davis, K. A., Fisher, V. A., Slopen, N., Smith, A. D. A. C., Tiemeier, H., & Susser, E. S. (2018). What life course theoretical models best explain the relationship between exposure to childhood adversity and psychopathology symptoms: Recency, accumulation, or sensitive periods? *Psychological Medicine*, *48*(15), 2562–2572. https://doi.org/10.1017/S0033291718000181

Ehring, T. (2021). Thinking too much: Rumination and psychopathology. *World Psychiatry*, *20*(3), 441–442. https://doi.org/10.1002/wps.20910

Eisma, M. C., Epstude, K., Schut, H. A. W., Stroebe, M. S., Simion, A., & Boelen, P. A. (2021). Upward and downward counterfactual thought after loss: A multiwave controlled longitudinal study. *Behavior Therapy*, *52*(3), 577–593. https://doi.org/10.1016/j.beth.2020.07.007

Eisma, M. C., & Stroebe, M. S. (2017). Rumination following bereavement: An overview. *Bereavement Care*, *36*(2), 58–64. https://doi.org/10.1080/02682621.2017.1349291

Festinger, L. (1954). A theory of social communication processes. *Human Relations*, *7*, 117–140. https://doi.org/10.1177/001872675400700202

Gierk, B., Kohlmann, S., Kroenke, K., Spangenberg, L., Zenger, M., Brähler, E., & Löwe, B. (2014). The somatic symptom scale–8 (SSS-8): A brief measure of somatic symptom burden. *JAMA Internal Medicine*, *174*(3), 399–407.

Higgins, E. T. (1996). *Activation: Accessibility, and salience*. In E. T. Higgins, & A. W. Kruglanski (Eds.), *Social psychology: Handbook of basic principles* (pp. 133–168). The Guilford Press.

Hyland, P., Redican, E., Karatzias, T., & Shevlin, M. (2024). The International Grief Questionnaire (IGQ): A new measure of *ICD-11* prolonged grief disorder. *Journal of Traumatic Stress*, *37*, 141–153. https://doi.org/10.1002/jts.22986

Kahneman, D., & Miller, D. T. (1986). Norm theory: Comparing reality to its alternatives. *Psychological Review*, *93*, 136–153. https://doi.org/10.1037/0033-295X.93.2.136

Killikelly, C., & Maercker, A. (2017). Prolonged grief disorder for ICD-11: The primacy of clinical utility and international applicability. *European Journal of Psychotraumatology*, *8*(sup6), 1476441. https://doi.org/10.1080/20008198.2018.1476441

Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, *8*(1), 33267. https://doi.org/10.1525/collabra.33267

LaPlante, C. D., Hardt, M. M., Maciejewski, P. K., & Prigerson, H. G. (2024). State of the science: Psychotherapeutic interventions for prolonged grief disorder. *Behavior Therapy, Special Issue: State of the Science in Behavior Therapy: Taking Stock and Looking Forward*, *55*(6), 1303–1317. https://doi.org/10.1016/j.beth.2024.07.002

Löwe, B., Wahl, I., Rose, M., Spitzer, C., Glaesmer, H., Wingenfeld, K., Schneider, A., & Brähler, E. (2010). A 4-item measure of depression and anxiety: Validation and standardization of the Patient Health Questionnaire-4 (PHQ-4) in the general population. *Journal of Affective Disorders*, *122*(1), 86–95. https://doi.org/10.1016/j.jad.2009.06.019

Lundorff, M., Holmgren, H., Zachariae, R., Farver-Vestergaard, I., & O'Connor, M. (2017). Prevalence of prolonged grief disorder in adult bereavement: A systematic review and meta-analysis. *Journal of Affective Disorders*, *212*, 138–149. https://doi.org/10.1016/j.jad.2017.01.030

Malgaroli, M., Maccallum, F., & Bonanno, G. A. (2018). Symptoms of persistent complex bereavement disorder, depression, and PTSD in a conjugally bereaved sample: A network analysis. *Psychological Medicine*, *48*(14), 2439–2448. https://doi.org/10.1017/S0033291718001769

McEvoy, P. M., Watson, H., Watkins, E. R., & Nathan, P. (2013). The relationship between worry, rumination, and comorbidity: Evidence for repetitive negative thinking as a transdiagnostic construct. *Journal of Affective Disorders*, *151*(1), 313–320.

McLaughlin, K. A., Greif Green, J., Gruber, M. J., Sampson, N. A., Zaslavsky, A. M., & Kessler, R. C. (2012). Childhood adversities and first onset of psychiatric disorders in a national sample of US adolescents. *Archives of General Psychiatry*, *69*(11), 1151–1160. https://doi.org/10.1001/archgenpsychiatry.2011.2277

McLean, E., Singer, J., Laurita, E., Kahler, J., Levin, C., & Papa, A. (2022). Perception of grief responses: Are maladaptive grief responses and the stages of grief considered normal? *Death Studies*, *46*(6), 1414–1423. https://doi.org/10.1080/07481187.2021.1983890

Meyer, T. J., Miller, M. L., Metzger, R. L., & Borkovec, T. D. (1990). Development and validation of the penn state worry questionnaire. *Behaviour Research and Therapy*, *28*(6), 487–495.

Morina, N. (2011). Rumination and avoidance as predictors of prolonged grief, depression, and posttraumatic stress in female widowed survivors of war. *The Journal of Nervous and Mental Disease*, *199*(12), 921–927. https://doi.org/10.1097/NMD.0b013e3182392aae

Morina, N. (2021). Comparisons inform me who I am: A general comparative-processing model of self-perception. *Perspectives on Psychological Science*, *16*(6), 1281–1299. https://doi.org/10.1177/1745691620966788

Morina, N., & Schlechter, P. (2023). Habitual aversive and appetitive well-being comparisons in dysphoria: Introducing the comparison standards scale for well-being. *Journal of Affective Disorders*, *322*, 132–140. https://doi.org/10.1016/j.jad.2022.11.012

Nolen-Hoeksema, S. (1991). Responses to depression and their effects on the duration of depressive episodes. *Journal of Abnormal Psychology*, *100*(4), 569–577. https://doi.org/10.1037/0021-843X.100.4.569

Nolen-Hoeksema, S., Wisco, B. E., & Lyubomirsky, S. (2008). Rethinking rumination. *Perspectives on Psychological Science*, *3*(5), 400–424.

O'Connor, M., Vang, M. L., Shevlin, M., Elklit, A., Komischke-Konnerup, K. B., Lundorff, M., & Bryant, R. (2023). Development and validation of the Aarhus PGD scale for operationalizing ICD-11 and DSM-5-TR prolonged grief disorder. *Journal of Affective Disorders*, *342*, 201–209. https://doi.org/10.1016/j.jad.2023.09.022

Ottenbreit, N. D., & Dobson, K. S. (2004). Avoidance and depression: The construction of the cognitive–behavioral avoidance scale. *Behaviour Research and Therapy*, *42*(3), 293–313. https://doi.org/10.1016/S0005-7967(03)00140-2

Palan, S., & Schitter, C. (2018). Prolific.ac – A subject pool for online experiments. *Journal of Behavioral and Experimental Finance*, *17*, 22–27. https://doi.org/10.1016/j.jbef.2017.12.004

R Core Team. (2025). *R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna: Austria*.

Rosseel, Y. (2012). Lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*, 1–36. https://doi.org/10.18637/jss.v048.i02

Rueger, M. S., Steil, R., Lubik, S., Roll, S. C., & Lechner-Meichsner, F. (2024). A comprehensive investigation of comorbidities of prolonged grief disorder in a bereaved inpatient psychiatric sample. *Journal of Psychiatric Research*, *177*, 185–193. https://doi.org/10.1016/j.jpsychires.2024.07.014

Russell, D. (1996). UCLA loneliness Scale (Version 3): Reliability, validity, and factor structure. *Journal of Personality Assessment*, *66*(1), 20–40. https://doi.org/10.1207/s15327752jpa6601_2

Schlechter, P., Hellmann, J. H., & Morina, N. (2024). The longitudinal relationship between well-being comparisons and anxiety symptoms in the context of uncontrollability of worries and external locus of control: A two-wave study. *Anxiety, Stress, & Coping*, *37*, 602–614. https://doi.org/10.1080/10615806.2024.2306530

Schlechter, P., Hoppen, T. H., & Morina, N. (2024). Associations among posttraumatic stress disorder symptoms, life satisfaction, and well-being comparisons: A longitudinal investigation. *Journal of Traumatic Stress*, *37*, 448–459. https://doi.org/10.1002/jts.23018

Schlechter, P., König, M., McNally, R. J., & Morina, N. (2023). Crying over spilled milk? A network analysis of aversive well-being comparison, brooding rumination and depressive symptoms. *Journal of Affective Disorders*, *339*, 520–530. https://doi.org/10.1016/j.jad.2023.07.088

Schlechter, P., Lutz, N. M., Morina, N., Grant, J. E., Lochner, C., Chamberlain, S. R., Wilkinson, P. O., & Fritz, J. (2024). The Comprehensive Adversity Measure (CAM): A measure of early adversity and its severity. *Child Abuse & Neglect*, *158*, 107117. https://doi.org/10.1016/j.chiabu.2024.107117

Schlechter, P., Meyer, T., Hagen, M., Baranova, K., & Morina, N. (2025). Comparative thinking among university students: An ecological momentary assessment of upward comparisons, stress and learning behavior during exam preparation. *Social Psychology of Education*, *28*, 54. https://doi.org/10.1007/s11218-025-10026-8

Schlechter, P., Meyer, T., Kerneck, L. E., Schulze Mengering, P., Hoppen, T. H., König, M., & Morina, N. (2026). Comparative thinking and clinical social anxiety: Within- and between-person effects in a daily ecological momentary assessment study and a four-wave longitudinal study. *Clinical Psychological Science*, 21677026261424180. https://doi.org/10.1177/21677026261424180

Schlechter, P., & Morina, N. (2023). Aversive well-being comparisons in dysphoria and the role of brooding

rumination. *British Journal of Clinical Psychology*, *62*, 444–458. https://doi.org/10.1111/bjc.12418

Schoemann, A. M., Boulton, A. J., & Short, S. D. (2017). Determining power and sample size for simple and complex mediation models. *Social Psychological and Personality Science*, *8*(4), 379–386. https://doi.org/10.1177/1948550617715068

Schönbrodt, F. D., & Perugini, M. (2013). At what sample size do correlations stabilize? *Journal of Research in Personality*, *47*(5), 609–612. https://doi.org/10.1016/j.jrp.2013.05.009

Stroebe, M., & Schut, H. (1999). The dual process model of coping with bereavement: Rationale and description. *Death Studies*, *23*(3), 197–224. https://doi.org/10.1080/074811899201046

Taylor, S. E., Neter, E., & Wayment, H. A. (1995). Self-evaluation processes. *Personality and Social Psychology Bulletin*, *21*(12), 1278–1287. https://doi.org/10.1177/01461672952112005

Topper, M., Emmelkamp, P. M. G., Watkins, E., & Ehring, T. (2014). Development and assessment of brief versions of the Penn State Worry Questionnaire and the ruminative response scale. *British Journal of Clinical Psychology*, *53*(4), 402–421. https://doi.org/10.1111/bjc.12052

Treynor, W. (2003). Rumination reconsidered: A psychometric analysis. *Cognitive Therapy and Research*, *27*(3), 247–259. https://doi.org/10.1023/A:1023910315561

Vlaev, I., Chater, N., Stewart, N., & Brown, G. D. A. (2011). Does the brain calculate value? *Trends in Cognitive Sciences*, *15*(11), 546–554. https://doi.org/10.1016/j.tics.2011.09.008

Ward, M. K., & Meade, A. W. (2023). Dealing with careless responding in survey data: Prevention, identification, and recommended best practices. *Annual Review of Psychology*, *74*, 577–596. https://doi.org/10.1146/annurev-psych-040422-045007

Wolf, E. J., Harrington, K. M., Clark, S. L., & Miller, M. W. (2013). Sample size requirements for structural equation models: An evaluation of power, bias, and solution propriety. *Educational and Psychological Measurement*, *73*(6), 913–934. https://doi.org/10.1177/0013164413495237

World Health Organization. (2018). *International statistical classification of diseases and related health problems (11th Revision)*. https://www.who.int/classifications/icd/en/.

Yentes, R. D., & Wilhelm, F. (2021). *Careless: Procedure for computing indices of careless responding (Version 1.2.1)[Computer software]*.